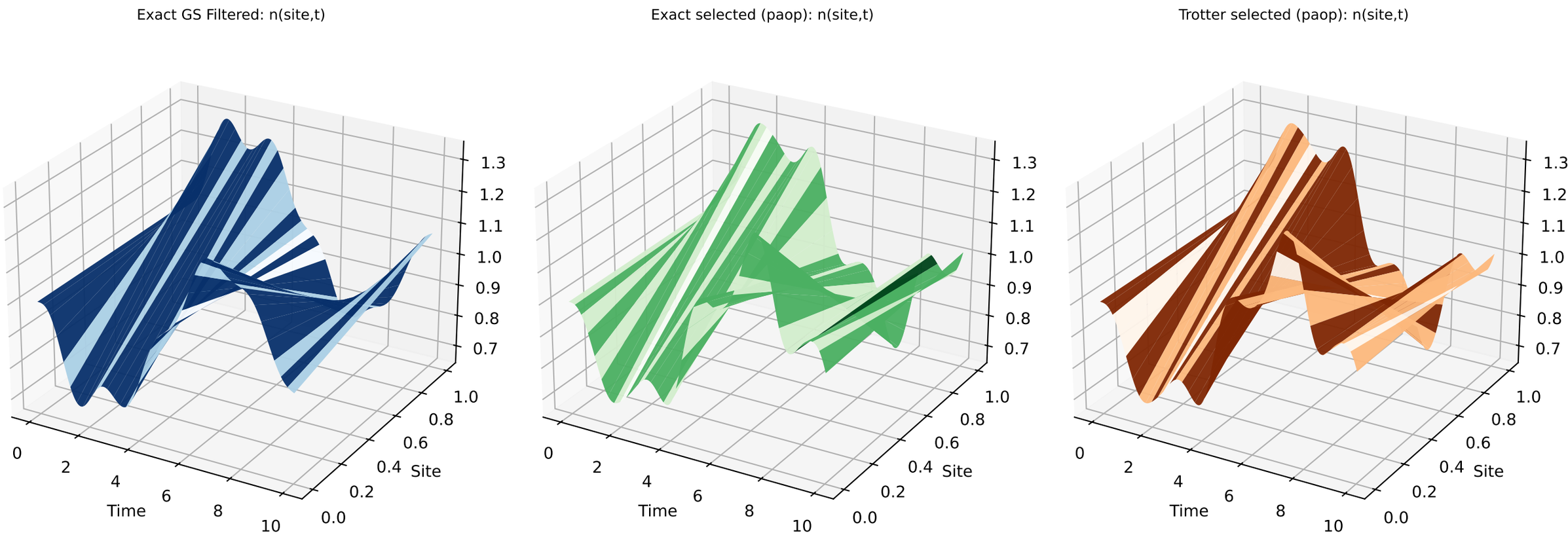
```
Parameter Manifest — Hubbard-Holstein L=2

Model:
  - model_family: Hubbard-Holstein
  - problem: hh

Ansatz:
  - ansatz_type: hh_hva_tw
  - vqe_method: hardcoded_hh_hva_termwise_statevector
  - initial_state_source: adapt_json
  - branch_order: exact_gs_filtered, exact_paop, trotter_paop, exact_hva, trotter_hva

Drive Enabled: True

Branch Provenance:
  - paop_source: adapt_json
  - paop_pool: paop_std
  - paop_depth: 11
  - hva_source: regular_vqe
  - hva_ansatz: hh_hva_tw
  - hva_vqe_success: True

ADAPT Import:
  - source_json: artifacts/json/adapt_hh_L2_J1_U2_g1_nph1_seed.json
  - metadata_match_passed: True
  - strict_match: True
  - pool: paop_std
  - ansatz_depth: 11
  - adapt_energy: -0.30277559216328104
  - adapt_abs_delta_e: 0.08677751113377519
  - A=0.5  omega=1.0
  - tbar=3.0  phi=0.0  t0=0.0
  - pattern=staggered  time_sampling=midpoint
  - exact_steps_multiplier=2
```

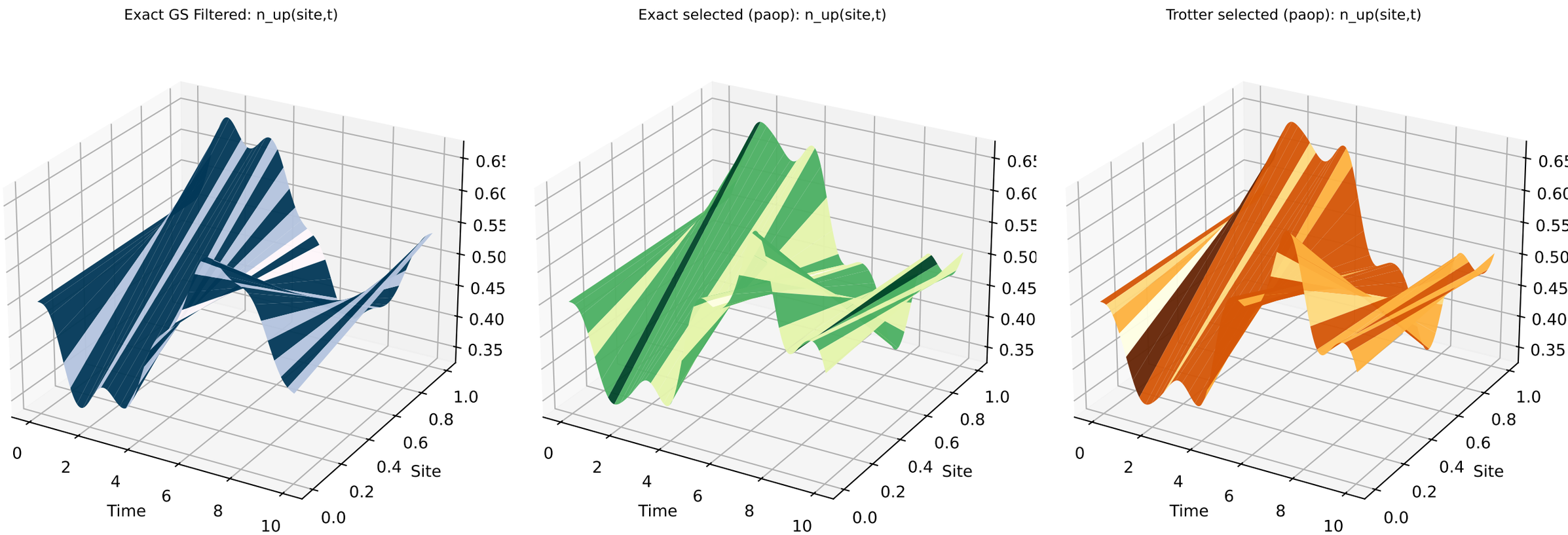
```
Core Physical Parameters:
  - t=1.0
  - U=2.0
  - dv=0.0

Hubbard-Holstein Parameters:
  - omega0=1.0
  - g_ep=1.0
  - n_ph_max=1
  - boson_encoding=binary
  - nq_fermion=4
  - nq_phonon=2
  - nq_total=6

Trajectory Settings:
  - t_final=10.0
  - num_times=201
  - trotter_steps=64
  - suzuki_order=2
  - term_order=sorted
  - boundary=periodic
  - ordering=blocked
  - initial_state_source=adapt_json

Topline Results:
  - subspace_fidelity_at_t0: 0.9755166487664396
  - vqe_energy: -0.3730234662832341
  - exact_filtered_energy: -0.38955310329705545
  - qpe_energy_estimate: None
```

```
ADAPT Initial-State Provenance

import_json: artifacts/json/adapt_hh_L2_J1_U2_g1_nph1_seed.json
import_source: adapt_vqe
strict_match: True
metadata_match_passed: True

pool_type: paop_std
ansatz_depth: 11
num_parameters: 11
operator_count: 11
adapt_energy: -0.30277559216328104
adapt_abs_delta_e: 0.08677751113377519

Dynamics used imported ADAPT state as t=0 initial condition.
```

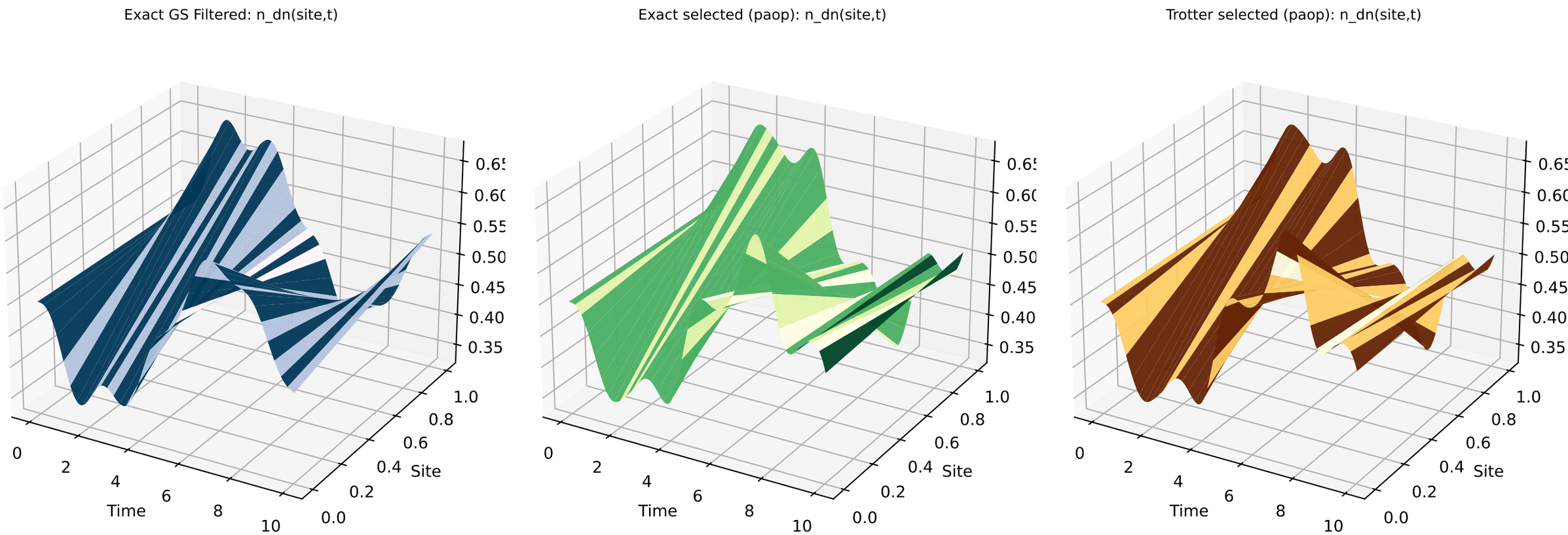
L=2 3D Densities (Total n)



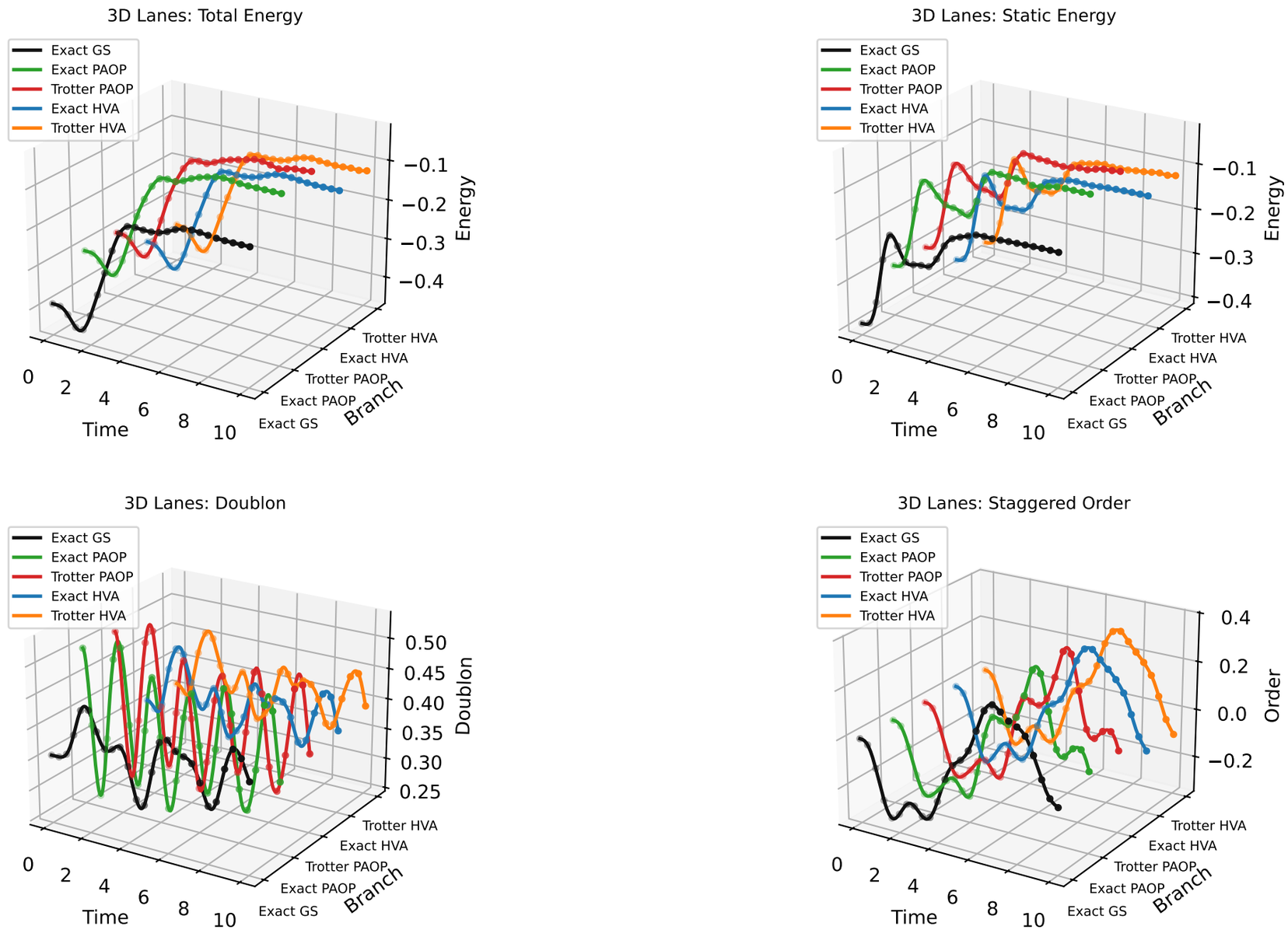
L=2 3D Densities (Spin-Up)

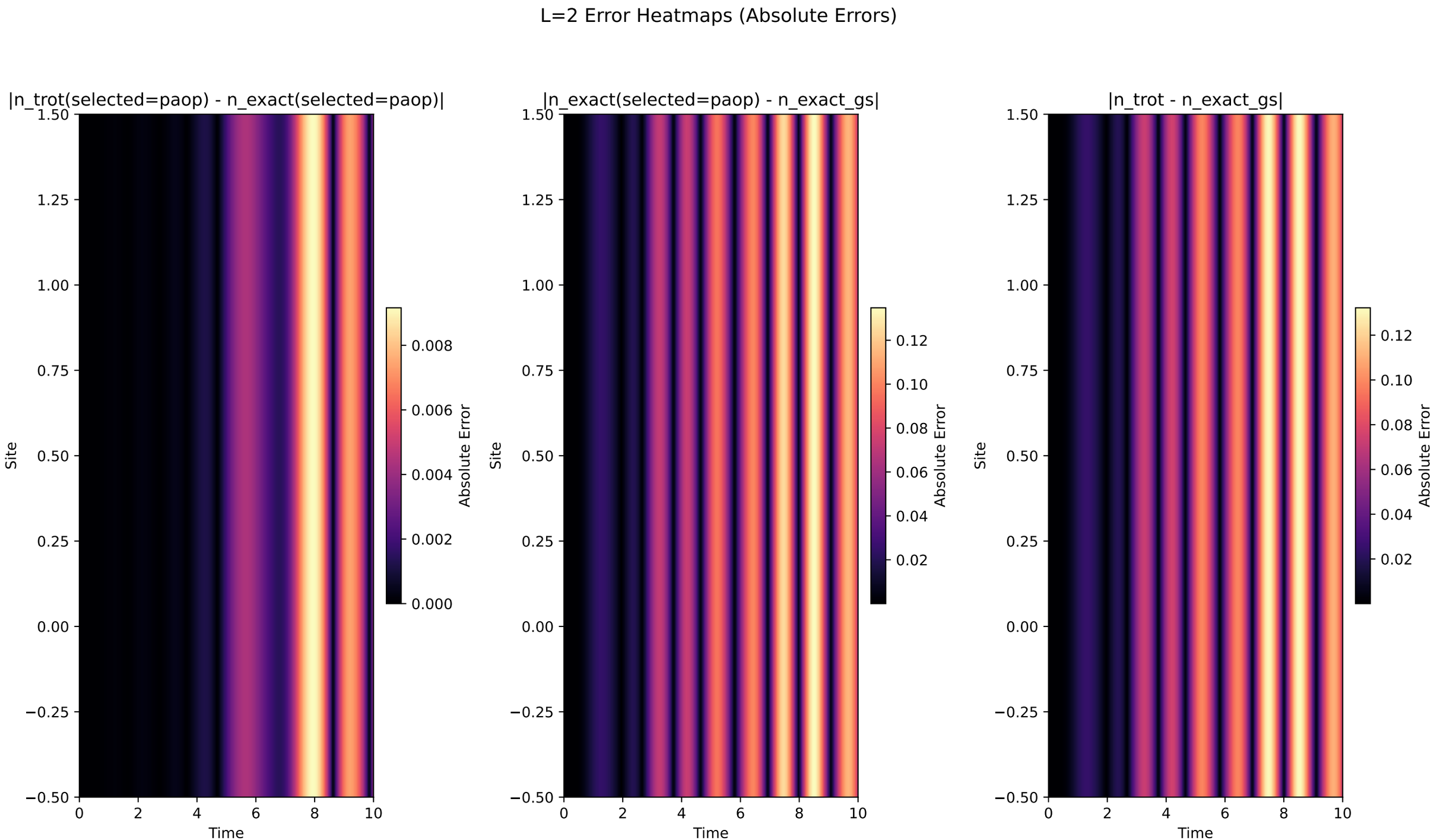


L=2 3D Densities (Spin-Down)

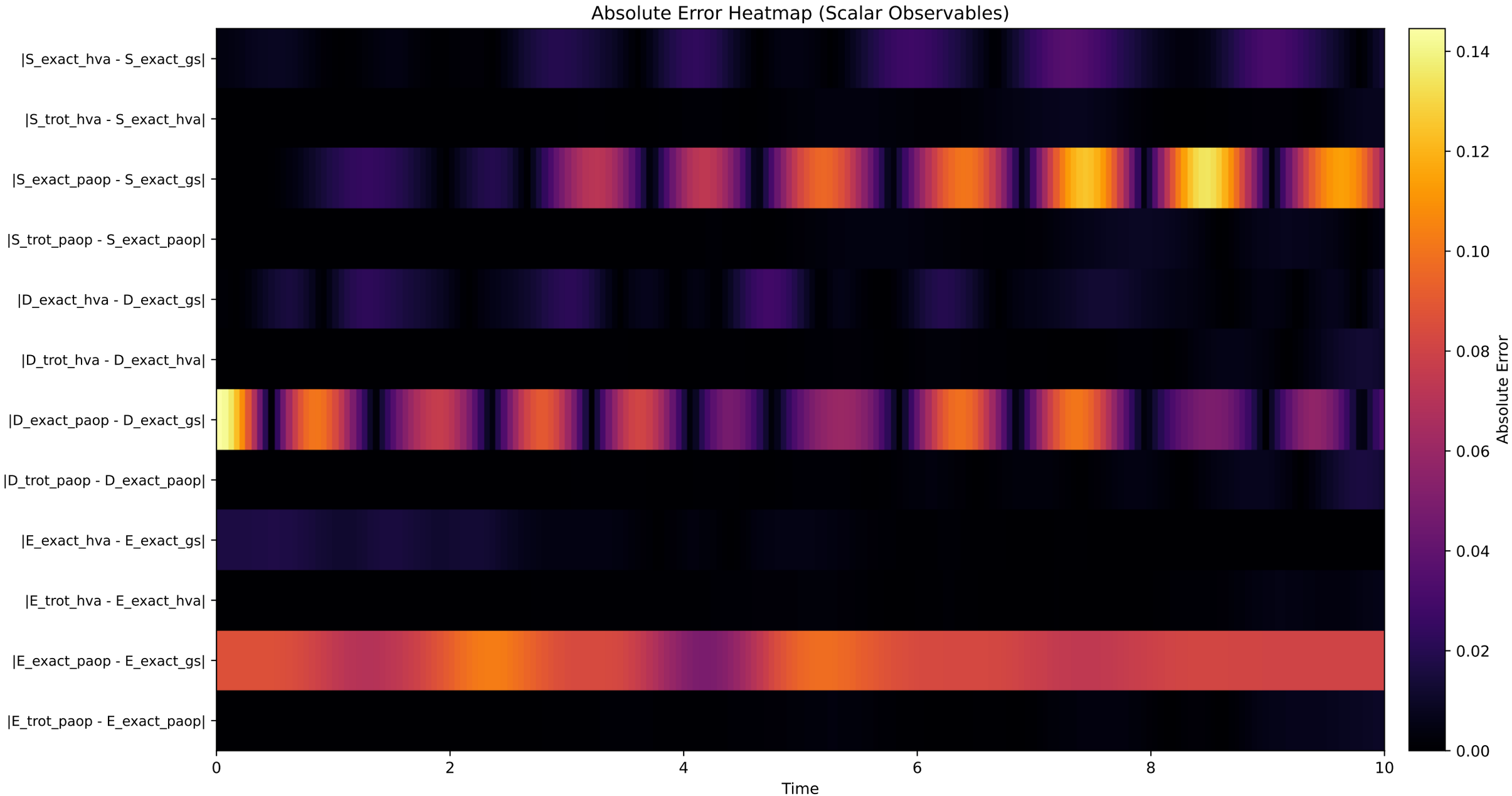


L=2 3D Scalar Observables (Five Branches)

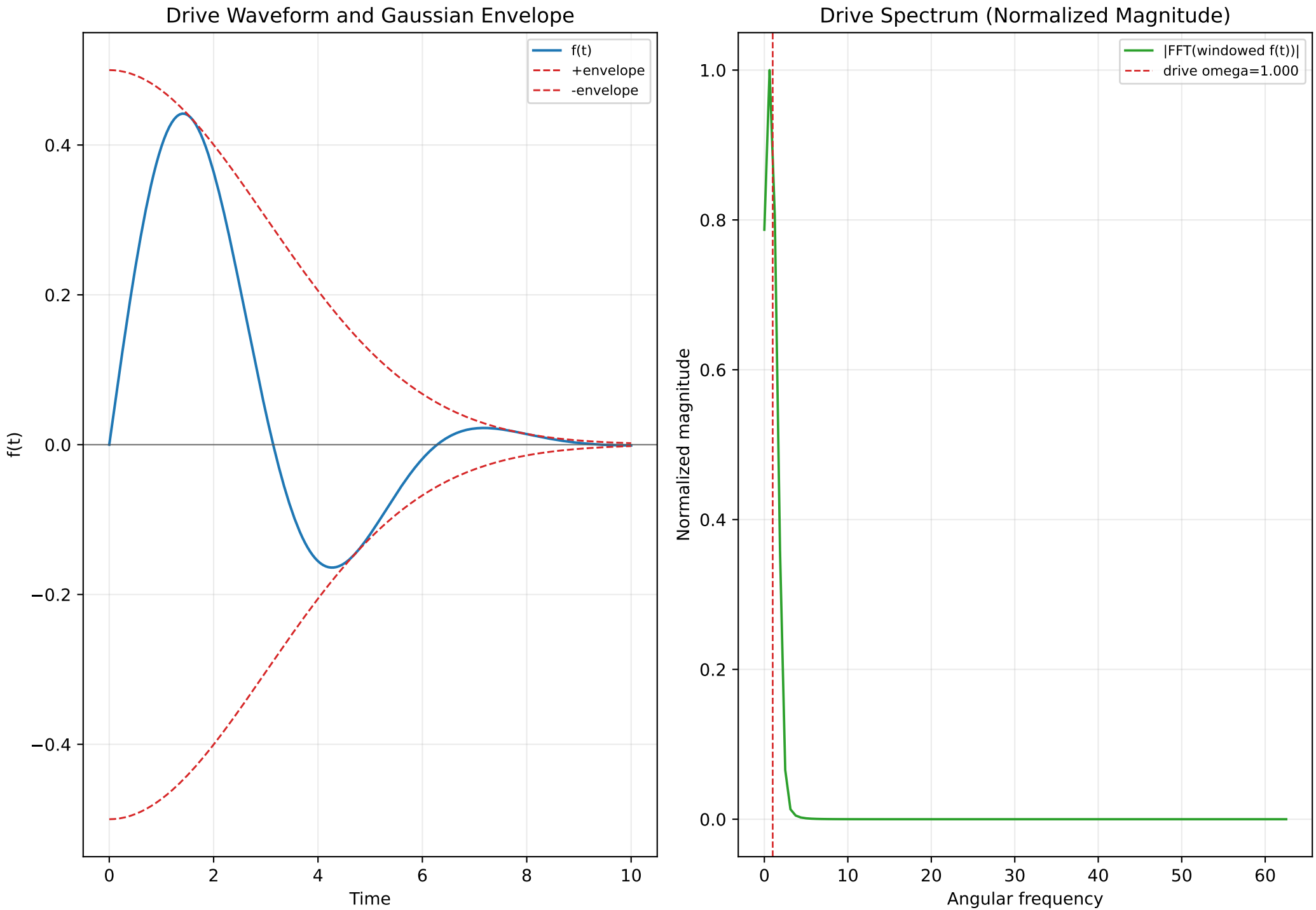



L=2 Error Heatmaps (Absolute Errors)
|n_trot(selected=paop) - n_exact(selected=paop)|
|n_exact(selected=paop) - n_exact_gs|
|n_trot - n_exact_gs|
Site
Time
Absolute Error
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
0
2
4
6
8
10
0.000
0.002
0.004
0.006
0.008
0.02
0.04
0.06
0.08
0.10
0.12

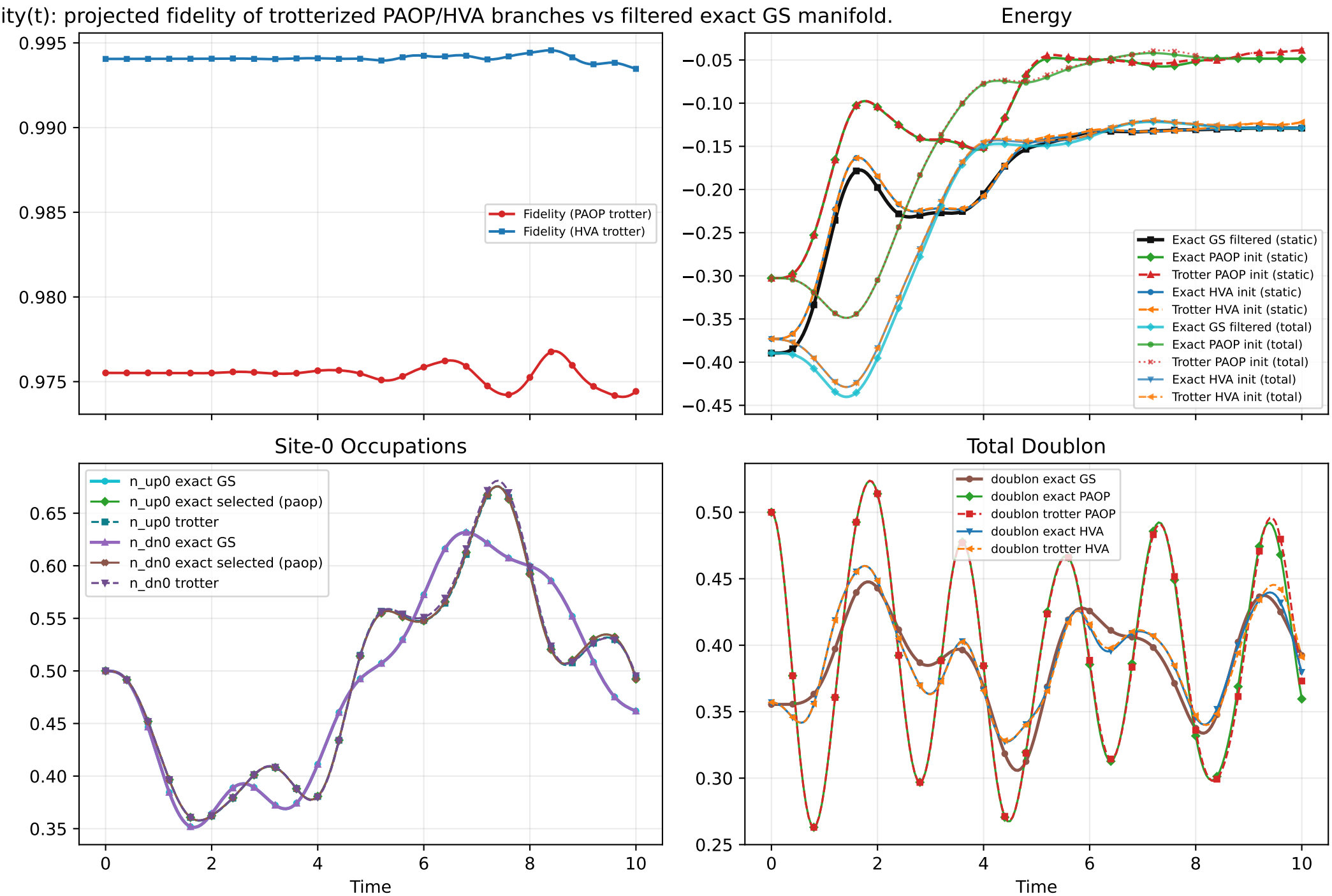
L=2 Scalar Error Heatmap

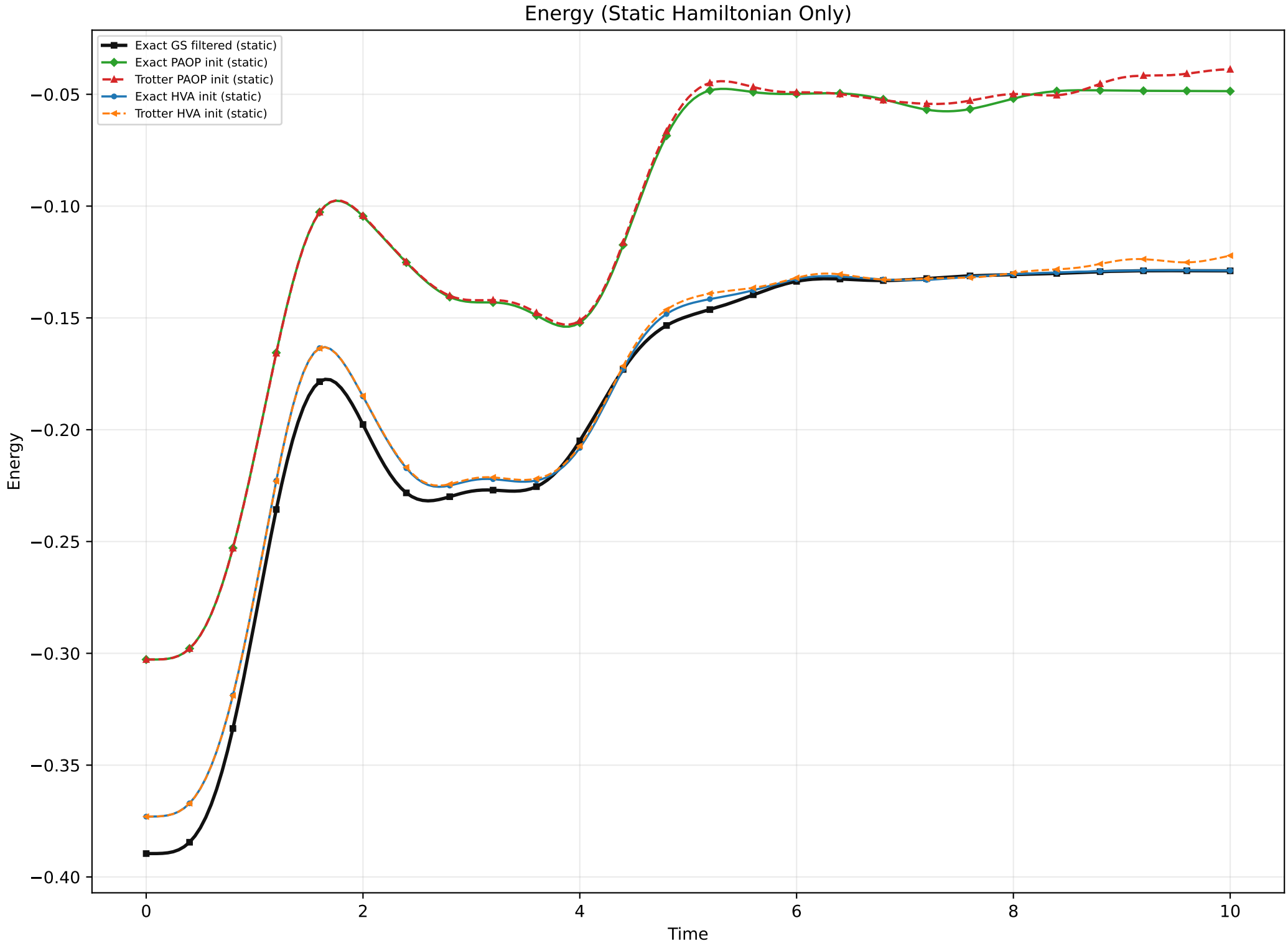


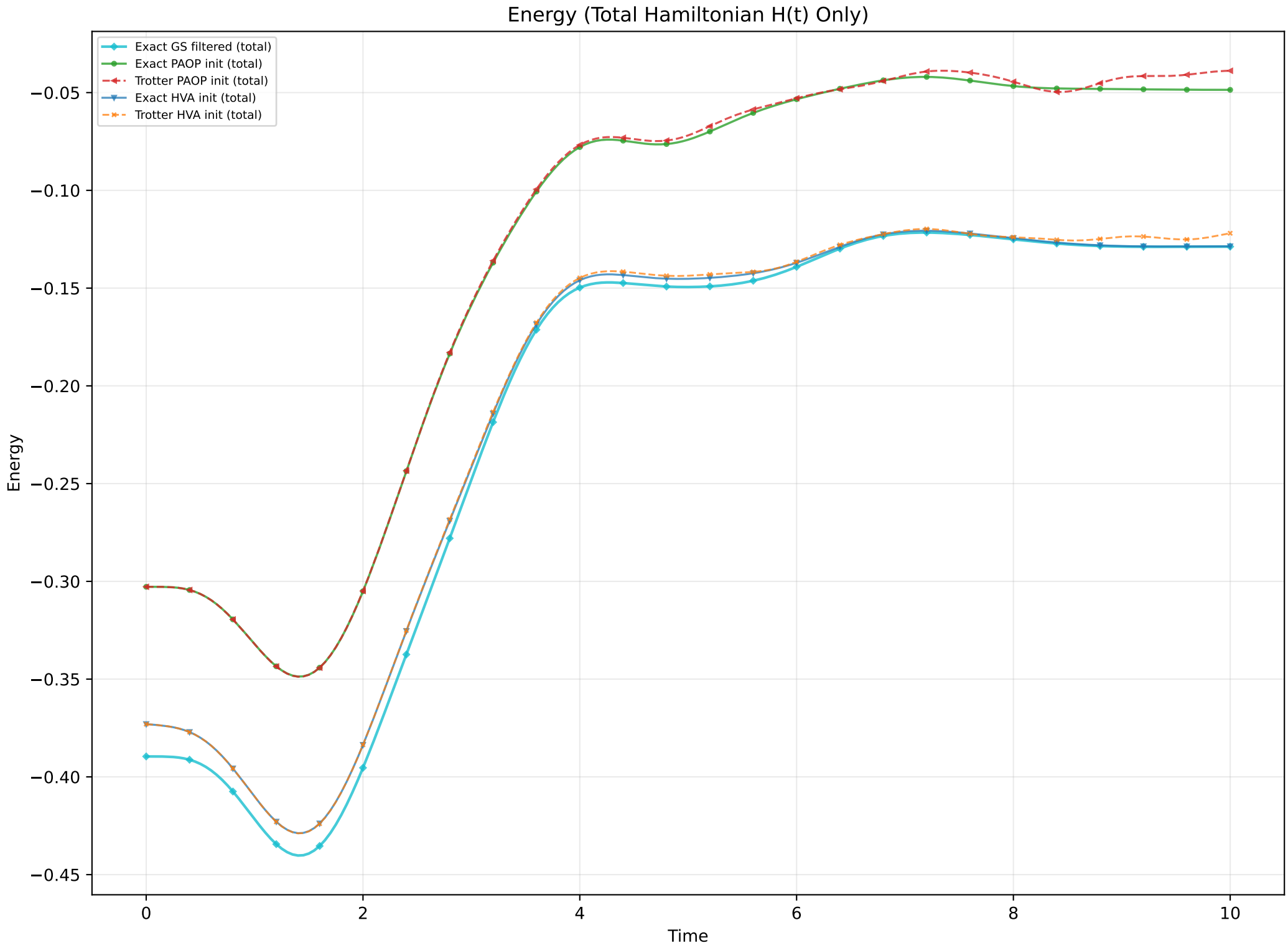
Drive Diagnostics: A=0.5, omega=1.0, tbar=3.0, phi=0.0, t0=0.0



Hardcoded Hubbard Pipeline: L=2



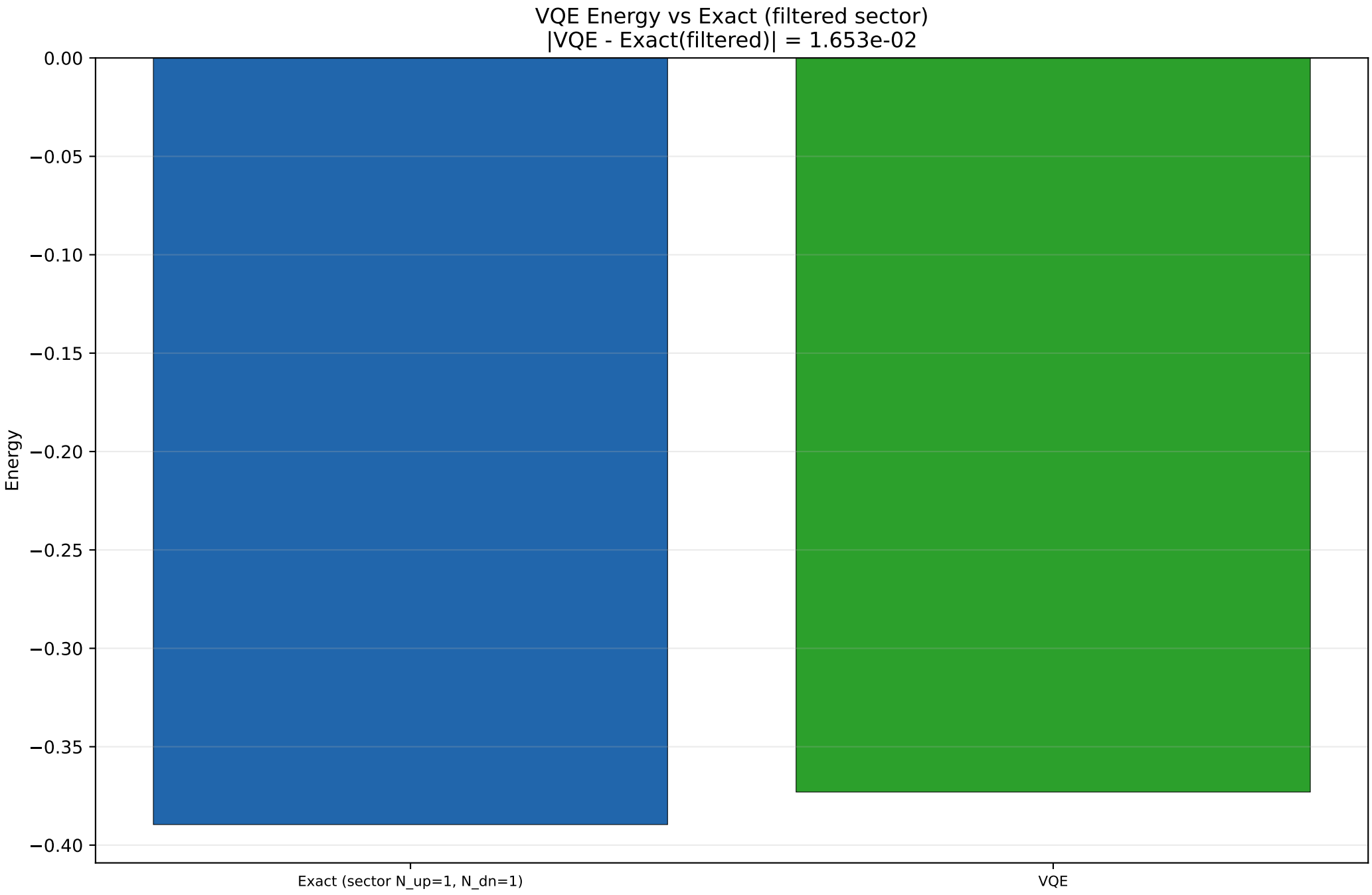

Energy (Static Hamiltonian Only)
Exact GS filtered (static)
Exact PAOP init (static)
Trotter PAOP init (static)
Exact HVA init (static)
Trotter HVA init (static)
Energy
Time
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
−0.35
−0.40
0
2
4
6
8
10


Energy (Total Hamiltonian H(t) Only)
Exact GS filtered (total)
Exact PAOP init (total)
Trotter PAOP init (total)
Exact HVA init (total)
Trotter HVA init (total)
Energy
Time
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
−0.35
−0.40
−0.45
0
2
4
6
8
10

Appendix — L=2

This section contains supporting/redundant views and metadata.
Core non-redundant analysis appears earlier in the PDF.

VQE optimises within the half-filled sector; exact (filtered) is the true sector ground state.
Full-Hilbert exact energy is in the JSON text summary only.



```
Hardcoded Hubbard pipeline summary

Ansatz:
  - hardcoded ansatz: hh_hva_tw
  - vqe method: hardcoded_hh_hva_termwise_statevector

Energy + Fidelity:
  - subspace_fidelity_at_t0: 0.9755166487664396
  - subspace_fidelity_paop_t0: 0.9755166487664396
  - subspace_fidelity_hva_t0: 0.994053547720396
  - energy_t0_exact_gs: -0.3895531032970548
  - energy_t0_exact_paop: -0.30277559216328137
  - energy_t0_trotter_paop: -0.30277559216328137
  - energy_t0_exact_hva: -0.37302346628323657
  - energy_t0_trotter_hva: -0.37302346628323657
  - ground_state_exact_energy_full_hilbert: -0.495507656605
  - ground_state_exact_energy_filtered: -0.38955310329705545
  - filtered_sector: {'n_up': 1, 'n_dn': 1}
  - vqe_energy: -0.3730234662832341
  - qpe_energy_estimate: None

Config:
  - initial_state_source: adapt_json
  - exact_trajectory_label: Exact_Hardcode
  - exact_trajectory_method: python_matrix_eigendecomposition
  - fidelity_definition: fidelity_paop_trotter(t) = <psi_paop_trot(t)|P_exact_gs_subspace(t)|psi_paop_trot(t)> and
    fidelity_hva_trotter(t) = <psi_hva_trot(t)|P_exact_gs_subspace(t)|psi_hva_trot(t)>, where
    P_exact_gs_subspace(t) projects onto the time-evolved filtered-sector ground manifold selected by E <= E0 +
    tol. Legacy key fidelity(t) follows the selected initial-state branch.
  - hamiltonian_terms: 17
  - reference_sanity: {'checked': False, 'reason': 'no matching bundled reference for these settings'}
```